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To the Editors,
*Pattern Recognition*

Re: "WAGER: An Exact Decomposition of Model-Pair Score Gains on Benchmarks with Strong Label Priors" — new submission

Dear Editors,

Please consider the enclosed manuscript for publication in *Pattern Recognition*.

**Why the content matters.** A scene graph generation method reports a ten-point gain in mean recall; a long-tailed classifier reports two points of accuracy. On benchmarks whose labels are partly determined by a feature both compared models observe — the object-class pair in relation prediction, the class frequency in long-tailed recognition — such a gain mixes two distinct improvements. The new model may fit the label frequencies *within* groups of that feature more closely, which a frequency table also achieves and which a shift in those frequencies removes; or it may match its predictions to the individual cases they were made for. Mean recall, zero-shot recall, GQA-OOD-style stratification and significance tests all leave the two mixed together, because every one of them slices a single model's performance rather than attributing the gain between two systems.

This paper decomposes that gain exactly. Score each model's prediction for one test case against the label of a *different* case in the same group: averaging over such within-group relabellings gives the part of the gain that survives, and what remains is a within-group covariance between the two models' predicted-probability difference and the label indicator. The identity is exact in finite samples, needs no fitted baseline model, no held-out fold and no tuning parameter, and costs $O(NK)$ on cached predictions, so any pair of frozen models can be audited from released probability outputs. It comes with a one-pass influence function giving image-cluster-robust intervals, a group-stratified randomization test, and a limit theorem.

The result we would put first is an audit of somebody else's published method. Applied to the two publicly released MOTIFS checkpoints of Tang et al. (CVPR 2020) on the canonical VG150 PredCls split, evaluated with the original authors' own code, the decomposition shows that TDE's ten-point mean-recall improvement corresponds to a proper-score difference that is overwhelmingly of the group-frequency kind, with a case-level remainder at least an order of magnitude smaller under either of two proper scores. Since TDE's construction subtracts a context-only prediction, this characterises the method rather than indicting it — but it does mean a metric the field reads as evidence about rare predicates can move ten points while the quantity that would measure case-level evidence barely moves.

**Why it fits this journal.** The contribution is a methodological one about how pattern-recognition systems are compared, grounded in the pattern-recognition literature on dataset bias, scene graph debiasing, long-tailed recognition and prior-aware evaluation metrics, and validated on the public benchmarks that literature uses. It is not a routine application of existing methods to a new dataset: the decomposition, its exactness, and the inference attached to it are new. The audience is the reader who has to decide what a leaderboard improvement means.

**Was your study compared to the state of the art?** Not in the usual sense, and we want to be straightforward about why: this is an evaluation method, not a predictor, so there is no accuracy to beat and no leaderboard position to claim. What it can be compared against is the existing practice for the same question — how the field currently tries to tell prior-fitting from case-level improvement — and the paper does that throughout Section 3. The five works that represent that state of the art, and that we position against directly, are: Zellers et al. (2018), whose frequency baseline is the standard prior-only reference; Tang et al. (2020), whose mean-recall reporting and causal intervention are the dominant debiasing practice in scene graph generation and whose released checkpoints we audit; Li et al. (2022), on the pathologies of the mean-recall metric itself; Kervadec et al. (2021), whose GQA-OOD stratifies evaluation by answer rarity within question groups; and Lu et al. (2016), which introduced zero-shot recall. Each is a prior-aware slicing, split or stress test of a *single* model. None decomposes the gain between two fixed systems with a finite-sample identity and uncertainty, which is what we add.

**Which public datasets are used?** All experiments use public benchmarks and, where possible, publicly released model weights. The main study uses Visual Genome via the canonical VG150 PredCls split, with the two released MOTIFS/MOTIFS-TDE checkpoints of Tang et al. evaluated using the authors' own code; a second study on the same benchmark uses predictors we train ourselves, including one built on a frozen CLIP ViT-B/32 encoder reading real image crops. Two further studies, reported in the appendices, use CIFAR-100-LT across seeds and imbalance ratios, and a long-tailed 20-Newsgroups split. No proprietary or restricted data is used. All code, cached model outputs and analysis scripts are released, including a verification script that checks each of the 89 reported numerical values against the committed results files.

**Which validation measures are used?** Because the object is a measuring instrument rather than a predictor, validation is of the estimator itself and uses four kinds of measure. First, recovery and calibration under known ground truth: monotone recovery of injected within-group signal, and type-I error of the randomization test (0.030 at level 0.05 over 200 null datasets). Second, coverage of the primary interval: 94.0% against a nominal 95% in an image-clustered, heavy-tailed regime, against 88.6% for an interval that wrongly assumes independence. Third, discriminant validity: designs establishing that prior-only improvements stay out of the case-level channel, that unrecorded within-group shortcuts enter it, and that monotone recalibration moves score between the channels — the last being a genuine limitation, for which the paper adopts a confidence-matching protocol and shows it works. Fourth, on real data, the conventional benchmark measures are reported alongside the decomposition so the two can be read against each other: top-1 accuracy, recall@50, mean recall@50 and zero-shot recall@50 for scene graph generation, and accuracy by frequency tier for the long-tailed studies.

**Declarations.** The manuscript is original, has not been published previously, and is not under consideration elsewhere. An earlier and substantially different version was submitted to a statistics journal and declined; that submission is closed. I am the sole author and approve this submission, declare no competing financial or personal interests, and received no specific grant for this work. The manuscript includes a CRediT statement and a declaration of generative AI use in its preparation, in accordance with journal policy. Highlights are supplied as a separate file.

Thank you for your consideration.

Sincerely,
Quang-Vinh Dang